Akash Prajapati

Generative AI Engineer | RAG, LLM Applications & Python Backend

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PROFESSIONAL SUMMARY

Generative AI / Python Backend Engineer with 3.3+ years of experience developing scalable backend systems and AI-powered applications. Hands-on experience building end-to-end RAG pipelines, LLM-powered workflows, semantic and multimodal retrieval, and voice-enabled AI applications using Python, Django, LangGraph, Sentence Transformers, ChromaDB and OpenAI APIs. Experienced in document processing, embedding generation, retrieval optimization, LLM evaluation, AI safety controls, tool calling and observability, with a strong foundation in developing and integrating production-oriented backend services.

TECHNICAL SKILLS

Programming & Backend: Python, SQL, Django, Django REST Framework, FastAPI, REST APIs, Microservices

Applied AI & GenAI: LLMs, RAG, Prompt Engineering, OpenAI APIs, LangChain, LangGraph, LangSmith, LLM Evaluation, Guardrails, Response Validation, Semantic Retrieval, Sentence Transformers, Text Embeddings, ChromaDB, Whisper, ElevenLabs, Image Embeddings

Databases & Messaging: PostgreSQL, Redis, Django ORM, Celery, Apache Kafka

Authentication & Integrations: Keycloak, JWT, OAuth 2.0, OTP Verification, Payment Gateway Integration, Third-Party API Integration, RBAC

Concepts: OOP, Database Design, Query Optimization, System Design, Agile Methodology

Cloud & Tools: AWS (S3, RDS, EC2, ELB), Docker, Kubernetes (Fundamentals), Grafana, Git, GitHub, Swagger/OpenAPI

PROFESSIONAL EXPERIENCE

Vishleshan Software Solutions Pvt. Ltd. | Software Developer 25th November 2025 – 24th September 2026

- Developed backend APIs and an end-to-end Retrieval-Augmented Generation (RAG) pipeline using Python and Django to generate context-grounded responses from a product knowledge base.
- Implemented document ingestion and recursive chunking using 500-word chunks with 20-word overlap; generated Sentence Transformers embeddings and stored vector representations in ChromaDB for approximately 100–500 knowledge-base items.
- Developed semantic retrieval using cosine similarity and dynamic similarity thresholds, with LangGraph conditional workflows for retrieval, response generation, query rephrasing and fallback handling.
- Integrated Whisper, GPT-4 and ElevenLabs to build a Hindi voice-enabled AI application covering speech-to-text, LLM-based response generation and text-to-speech.
- Implemented LLM reliability and safety controls, including PII redaction, structured response validation and prompt-injection detection; integrated LangSmith for execution tracing, debugging and latency/token monitoring.
- Evaluated RAG response quality across approximately 5,000 test cases using expected answers and LLM-based evaluation, achieving an average evaluation score of approximately 87%.
- Implemented LLM tool-calling workflows to trigger backend API operations and email-sending functionality, connecting AI-generated decisions with application services.
- Developed AI-powered Smart Search using separate image and text embedding pipelines, enabling image, part-number, product-name, category and voice-based discovery with top-5 similarity retrieval; containerized the application using Docker and deployed it on AWS EC2.
- Developed 100+ REST APIs using Python, Django and DRF for enterprise e-commerce workflows supporting 1,000+ daily orders; built FastAPI notification services integrated with Apache Kafka, handling approximately 1,000–10,000 notifications daily.
- Implemented Redis caching and Django ORM query optimizations; developed asynchronous reporting and export workflows using Celery and Redis, AWS S3 storage and secure presigned downloads, alongside DSR/MSR, Shiprocket and SAP Grow integrations.

Sankey Solutions | Solution Analyst 1st June 2023 – 14th November 2025

- Independently developed an AI-assisted Git repository analysis solution, integrating GitHub APIs to retrieve commit differences and surrounding code context for automated change analysis.
- Implemented separate OpenAI API workflows for code-change summarization and potential security vulnerability detection, generating structured findings with issue descriptions and severity levels.
- Developed security checks to identify exposed API keys, access tokens, passwords, email credentials and AWS access details in repository changes.
- Developed backend services across vehicle management, subscriptions, payments and fleet operations within an approximately 18-microservice architecture using Python, Django and PostgreSQL.
- Built a Vehicle Day-wise Tracker backend, processing historical GPS coordinates, calculating distance using the Haversine formula, identifying trips and stops, and preparing route data for Google Maps visualization.
- Improved API performance by approximately 20–30% by resolving N+1 query issues and optimizing database access using Django ORM Subquery and related query optimizations.
- Implemented role- and subscription-based access controls, managing Fleet Owner and Fleet Manager permissions, site-level vehicle visibility and plan-based access to fleet-management features.

- Developed mileage analytics and historical telemetry synchronization workflows, processing driving parameters and integrating third-party APIs through scheduled incremental fetching, checkpoint-based recovery and PostgreSQL upserts.
- Integrated CCAvenue payment gateway workflows, including request encryption, response decryption, payment callbacks, status updates and subscription synchronization; developed purchase, upgrade, renewal and expiry workflows.
- Developed a quality-management backend using Django and PostgreSQL, implementing 50+ REST APIs, Maker-Checker inspection workflows and an Apache Kafka consumer for manufacturing vehicle data ingestion.

EDUCATION

Bachelor of Engineering — Computer Engineering
Smt. Indira Gandhi Engineering College, University of Mumbai | Aug 2019 – Jun 2023 | CGPA: 8.88/10

ACHIEVEMENTS

- **Shining Star:** Recognized for outstanding technical contributions.
- **Bar Raiser:** Recognized for problem-solving, technical guidance, and supporting team members.

CERTIFICATIONS

- Machine Learning with Google Cloud
- Android Development

PUBLICATION

- Crop Protection Against Birds Using Deep Learning and IoT, IRJET, Vol. 10, Issue 4, April 2023.